

Nithin Kumar Reddy

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SUMMARY

CS undergrad building production-oriented AI systems: multi-agent orchestration, cloud data pipelines (Azure/Databricks), and autonomous developer tooling (AST, LSP, RAG). Backend focus (FastAPI, ETL, monitoring) with open-source cloud security contributions (Prowler). Seeking SWE / platform / AI-infra internships.

EDUCATION

PES University

B.Tech, Computer Science and Engineering

Coursework: DSA, OOAD, DBMS, OS, Networks, Compilers, Cloud Computing, ML, Big Data, Data Analysis

Aug 2023 -- May 2027

Bengaluru, Karnataka

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, Bash

Backend: FastAPI, Node.js, Express, REST, WebSockets, JWT/OAuth

Cloud & Data: Azure (ADF, Functions, ADLS Gen2), Databricks, Spark, Kafka, AWS, Docker, Kubernetes, Linux, Git

AI/ML: Multi-agent systems, RAG, LangChain, HF Transformers, scikit-learn, TensorFlow, OpenCV, Pandas/NumPy

Frontend & DBs: React, Playwright, FFmpeg; PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Supabase

EXPERIENCE

IFSCR (Center for Information Security, Forensics and Cyber Resilience)

Jun 2025 -- Aug 2025

Summer Research Intern -- Digital Image Forensics

PES University

- Built an image provenance pipeline to identify original sources of manipulated/reposted images (feature matching + temporal graph analysis).
- Implemented SIFT matching resilient to scale/crop/rotation; fused similarity + graph signals into confidence-ranked source prediction.
- Delivered a research prototype with reproducible evaluation for multi-hop image reuse scenarios.

PROJECTS

Open Source -- Prowler (prowler-cloud/prowler)

2025 -- 2026

Python, AWS, cloud security compliance

- Contributions to Prowler (open-source cloud security): PR #12076 EC2 stopped-instance check (github.com/prowler-cloud/prowler/pull/12076); PR #12118 SageMaker endpoint KMS encryption with service model + tests (github.com/prowler-cloud/prowler/pull/12118).

Aegis Engineer

2025 -- 2026

Python, multi-agent systems, AST, LSP, semantic search

- Built an autonomous software-engineering platform with repo intelligence (AST/LSP/semantic search) and multi-agent planning for reviewable, PR-style fixes. github.com/Nithin078/Aegis-Engineer

Agentic Orchestration for Big Data Pipelines

Jan 2025 -- May 2025

Python, FastAPI, Azure Data Factory, Databricks, React, scikit-learn

- Architected a 7-agent ETL orchestrator on ADF + Databricks; ML compute sizing cut over-provisioning ~30%; monitoring + mid-run reallocation cut simulated failures ~25%.
- Dual-validation Assurance Agent (structural + LLM semantic checks) + React analytics dashboard via FastAPI.

DemoGen

Mar 2026 -- Apr 2026

Python, FastAPI, React, Playwright, FFmpeg, TTS

- Built an AI demo-video platform: LLM-planned Playwright automation, screen capture, TTS narration, and FFmpeg composition with FastAPI + React control plane. github.com/Nithin078/DemoGen

ACHIEVEMENTS & CERTIFICATIONS

- Top 4 -- FASTATHON (SRM All-India) | Top 4 -- Space Race Hackathon (PES) | Adobe India Hackathon
- Oracle Cloud Infrastructure -- AI Foundations Associate | HackerRank -- Problem Solving (Intermediate)